

$$\begin{array}{ccc}
[B, A]_D \otimes_D B & \xlongequal{\quad} & F(\iota[B, A]_D \otimes_C \iota B) \\
\downarrow \text{ev}_A^B & & \downarrow F(\psi_{B,A} \otimes_C \iota B) \\
& & F([\iota B, \iota A]_C \otimes_C \iota B) \\
& & \downarrow F(\text{ev}_{\iota A}^{\iota B}) \\
A & \xleftarrow{\quad \epsilon_A \quad} & F\iota A
\end{array}$$